

**FITNESSWISE**

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STORIES

Faster, higher, stronger

How motion, energy, and force **combine to power** peak athletic performance.

We all love sports, don't we? It does not require an IQ of 200+ to know that athletes want to improve their performance, prevent injuries, and gain an edge over their opponents. In that case, a good knowledge of science can be of great help in understanding various aspects of physiology and the physics related to force, matter, energy and motion. It helps in understanding how things work — from the smallest finer movement to the largest or gross movement patterns.

Physics can also help athletes master their respective sports, be it cricket or Formula One. In that parlance, we will explore some of the physics concepts that are applicable to sports, and how you can use them to your advantage.

The power of force

One of the most important physics concepts in sports is force. Force is a push or a pull that causes an object to change its speed, shape, or direction. It can be applied in various ways — through muscles, gravity, air resistance, and friction. It can also be measured by its magnitude (how strong it is) and its direction (which way it points).

In sports, force affects the speed of movement, farthest distance thrown, height of jumps, and intensity of impact.

According to the Third Law of Motion, when you run, you apply a force to the ground with your feet, and the ground applies an equal and opposite force back to you. This ground reaction force propels you forward and upward, allowing you to run faster and longer. However, if the ground is too soft or slippery,

the force is diminished, and you lose speed and stability. A simple example is running on grass as compared to running on a beach.

Similarly, when you throw a ball, you apply a force to the ball with your arm and hand, and the ball applies an equal and opposite force back to you. The ball's force pushes you slightly backwards, while it flies forward with a certain speed and direction. However, if the ball is too heavy or too light, the force will vary, and the ball will not go as far or as fast as you want.

To use force effectively in sports, you need to know how much force to apply, in which direction, and for how long. You also need to consider the effects of other forces acting on you or the object you are moving — such as gravity, air resistance, friction or wind direction. These forces will always be present to varying degrees. By adjusting your technique and training accordingly, you can optimise your performance and avoid injuries.

The art of work and energy

Another key physics concept in sports is work and energy.

In physics, work is defined as the energy transferred to or from an object by a force acting upon it as it is displaced. It is done when a force causes a displacement of an object. Energy, however, is the ability to do work. Some forms of energy are specific to sports and closely related in their transfers:

Kinetic energy: Energy of motion

Potential energy: Stored energy that has the potential to do work (e.g., a stretched rubber band, a ball held above the ground)

Thermal energy: Energy associated with temperature and heat